Dark Patterns Data Mining Project Report

A two-dataset machine learning study using separate training and testing data

1. Introduction

This project studies dark patterns in e-commerce text. Dark patterns are messages or interface designs that try to push users into actions they may not fully want, such as buying quickly or making a rushed decision.

The goal of this project is to use Python and machine learning to detect whether a text is a dark pattern or not a dark pattern by training on two combined datasets.

2. Scope Of The Project

The project is limited to text-based detection. It does not analyse full website images, page layout, colors, or user behavior.

The main work is to clean two datasets, split them into training and testing sets, merge the split files, train multiple algorithms, compare their results, and choose the best one.

3. Dataset Used

The cleaned combined dataset contains 4219 text records after removing missing values and duplicate text-label pairs.

The original dataset contributes 2356 records and the synthetic dataset contributes 1863 records.

The binary labels are 1 for dark pattern and 0 for not dark pattern. In the merged cleaned data, there are 2694 dark-pattern rows and 1525 not-dark-pattern rows.

For this project, 3374 records were used for training and 845 records were used for testing.

4. Tools And Technologies

Programming language: Python

Main libraries: pandas, scikit-learn, joblib, openpyxl, reportlab

Text processing method: TF-IDF vectorization with unigram and bigram features

Algorithms tested: Linear SVM, Logistic Regression, SGD Log Loss, Complement Naive Bayes, and a hybrid stacking model

5. How The System Works

First, the project loads the original TSV dataset and the synthetic Excel dataset using the same text and label schema.

Next, each dataset is cleaned separately by selecting the needed columns, removing missing values, and removing duplicate text-label rows.

After that, each dataset is split into training data and testing data using an 80:20 ratio with a fixed random state. The original dataset produces 1884 training rows and 472 testing rows, while the synthetic dataset produces 1490 training rows and 373 testing rows.

Then, the two training splits are merged together and the two testing splits are merged together.

The merged training data is used to train each algorithm. The merged testing data is kept separate and used only for final evaluation.

Finally, the project compares the models using F1 score, accuracy, precision, and recall.

6. Pseudocode

```
START
Load original dataset
Load synthetic dataset

FOR each dataset
    Select shared columns
    Clean missing and duplicate text-label rows
    Split into training set and testing set
END FOR

Merge original and synthetic training data
Merge original and synthetic testing data

FOR each algorithm in [Linear SVM, Logistic Regression, SGD Log Loss, Complement Naive Bayes, Hybrid Stacki
    Convert training text into TF-IDF features
    Train algorithm using merged training data
    Predict labels on merged testing data
    Calculate accuracy, precision, recall, and F1 score
END FOR

Compare all results using F1 score first
Choose the best model
Save the best model and result files
END
```

7. Algorithms Used

Linear SVM is strong for text classification because it handles high-dimensional TF-IDF features very well.

Logistic Regression is a reliable binary classification model and gives interpretable linear decision boundaries.

SGD Log Loss is a fast linear classifier that works well on large sparse text features.

Complement Naive Bayes is a useful probabilistic baseline for imbalanced text data.

The hybrid stacking model combines Logistic Regression, a calibrated Linear SVM, and Complement Naive Bayes to test whether an ensemble can improve the final prediction.

8. Findings And Results

The following table shows the test results of the five algorithms on the merged testing data.

Algorithm	Accuracy	Precision	Recall	F1 Score
Hybrid Stacking	0.9657	0.9830	0.9630	0.9729
Linear SVM	0.9657	0.9867	0.9593	0.9728
SGD Log Loss	0.9645	0.9866	0.9574	0.9718
Logistic Regression	0.9633	0.9885	0.9537	0.9708
Complement NB	0.9479	0.9662	0.9519	0.9590

The best algorithm in this project is **Hybrid Stacking**. It achieved an F1 score of **0.9729**, an accuracy of **0.9657**, a precision of **0.9830**, and a recall of **0.9630** on the merged test set.

The best model is selected by F1 score first, then accuracy, and then precision if a tie happens.

9. Conclusion

This project shows that machine learning can be used to detect dark-pattern text with strong performance even when training data comes from two different sources.

Cleaning the datasets separately and merging the train and test splits gives a fairer evaluation than mixing duplicate rows across both stages.

Among the tested algorithms, Hybrid Stacking gave the best overall result on the merged dataset.

So, this project is suitable for a student data mining or machine learning study based on text classification with multiple datasets.

10. Future Improvement

More algorithms can be tested, such as transformer models like BERT or RoBERTa.

More text preprocessing can be tried, such as lemmatization, feature selection, or hyperparameter tuning.

The project can also be expanded from binary classification to multi-class classification using the pattern categories.

11. References

Dark Patterns dataset from Kaggle

Synthetic dark-pattern dataset used for external validation and additional training examples

Python libraries: pandas, scikit-learn, openpyxl, and reportlab documentation